AI Internship: Module 01 Learning Notes

Module 01: Introduction to AI & Python (Day 1 to Day 4, Beginner level)

This repository contains my learning notes and practice work for Module 01 of the AI internship.

Contents

File	Description
`Learning_Notes.pdf`	Consolidated learning notes for Module 01 (10 pages)
`README.md`	Project overview (this file)

Module Objectives

1. Learn the basics of Artificial Intelligence and Machine Learning.
2. Understand the difference between AI, ML and Deep Learning.
3. Explore real-world AI applications.
4. Install Python and practise programming concepts (variables, loops, functions, lists).
5. Write beginner-friendly Python programs.

What the Notes Cover

1. **Basics of AI & Machine Learning**: what AI and ML are, the three types of learning (supervised, unsupervised, reinforcement) and the ML workflow.
2. **AI vs ML vs Deep Learning**: how the three relate, a comparison table and a neural network overview.
3. **Real-World AI Applications**: healthcare, finance, retail, transport, education, agriculture and everyday tools, plus responsible AI.
4. **Python Setup & Core Concepts**: installation, variables and data types, operators, conditionals, loops, functions and lists.
5. **Beginner Python Programs**: even/odd, calculator, factorial, Fibonacci series, largest number in a list and palindrome check.
6. **Summary & Revision Checklist**: a quick recap of the module.

Deliverables (Module 01)

- [x] Learning Notes (`Learning_Notes.pdf`)
- [x] GitHub Repository (this repo)

Tech & Tools

- Python 3
- Editor: VS Code / PyCharm

Author

Jean Reno R

Modules 02 onwards will be added as they are completed.